Satellite Building Workshop

Space Faculty Private Limited

At Space Faculty, we chart pathways between earthly dreams to interstellar reality. We power brilliant and bold ideas in Space and frontier technologies through training, engineering, research and strategic public-private partnerships.

www.spacefaculty.asia

About Space Faculty Pte Ltd

Our expertise spans several key areas:

1. Satellite Development: From satellite building and launching to providing hands-on training in satellite development.

2. Scientific Advancement: Facilitating cutting-edge space experiments for the scientific community.

3. Workforce Development and Executive Programmes: Specialized Executive courses, training, workshops and hackathons to understand how space technologies can future proof your organization and your career

4. National-level Space and STEM 2.0 Roadmap Development: Executing impactful, roadmaps through training, "galaxy-class" competitions and hackathons with global experts to ignite youth interest in space and STEM fields.

HONORARY ADVISORS

- Prof Lui Pao Chuen, Mr Atsushi Taira, Mrs Koh Soo Boon, Mr Seah Moon Ming, Dr Victoria Coleman

CEO

- Ms Lynette Tan

Nelson Tan

- Education:
  - NTU BEng (ME)
- Space Faculty
  - Satellite Engineer

Rohini Parthasarathy

- Education:
  - Master of Space Studies, International Space University
  - B.E. (EEE), Chennai Institute of Technology
- Experience:
  - Space Faculty Pte Ltd : 2026 - present

Ang Eng Hian

- Education:
  - NUS BEng (EE)
  - NUS MSc (EE)
  - NUS MTech (SE)
- Space Faculty
  - Technical Consultant
- DSO National Laboratories
  - 2007 to 2025
  - Principal Engineer (Radar System Engineer, Satellite Payload System Engineer)
  - RF system, Electronic, Software, FPGA, Signal Processing

Lam Lap

- Education:
  - MSc in Electrical Engineering (UIUC, USA)
- Space Faculty
  - Technical Director
- DSO National Laboratories
  - 2003 to 2024
  - Programme Director

Agenda

S/N	Day	Time		Title	2nd Space Slides	Trainer	Duration
		Start	End				
1	1	9:00	9:30	INTRO Workshop agenda, hands-on objectives, Attendance, Final Exercise, SBJ, Team Building		NT	30
2		9:30	9:45	Satellite and Space Ecosystem		NT	15
3		9:45	10:00	Satellite & Space Mission	L1 Satellite and Space Mission	RHN	15
4		10:00	10:15	Satellite Payloads & Missions	L19 (part) Payloads and Examples of Satellite Mission	RHN	15
5		10:15	10:30	Break			15
6		10:30	11:30	Satellite Architecture, Design & Development Overview	L3 Satellite Sub-systems	RHN	60
7		11:30	12:00	[Lab] TinkerCad with arduino(Cubesat Digital Twin)	L11 Review and Programming (Workshop Materials), New	RHN	30
8		12:00	13:00	Lunch			60
9		13:00	13:30	[Lab] Cubesat hands on (inventory list chk, GS assembly)	**Slides Day1 (Arduino IDE, driver setup), Day2 (GS assembly, tes**	RHN, NT	30
10		13:30	15:00	[Lab] Cubesat hands on (Cubesat assembly)	**Slides Day3 (cubesat assembly)**	RHN, NT	90
11		15:00	15:15	Break			15
12		15:15	17:30	[Lab] Cubesat hands on (assembly, programming, initial tests)	**Slides Day3 (cubesat assembly), Day4 (OBC)**	RHN, NT	135

Agenda

S/N	Day	Time		Title	2nd Space Slides	Trainer	Duration
		Start	End				
13	2	9:00	10:15	Command & Data Handling Subsystem (CDHS)	L6 Satellite Command and Data Handling	RHN	75
14		10:15	10:30	Break			15
15		10:30	11:15	Electrical Power Subsystem (EPS)	L4 Electrical Power System	RHN	45
16		11:15	12:00	Thermal Control Subsystem (TCS)	L9 Satellite auxiliary systems_Thermal_Radiation	NT	45
17		12:00	13:00	Lunch			60
18		13:00	13:30	Radiation SEE	L9 Satellite auxiliary systems_Thermal_Radiation	RHN	30
19		13:30	14:00	Structures & cable harness	L5 Satellite Structures	NT	30
20		14:00	15:00	[Lab] Cubesat hands on (TM, LoRa)	**Slides Day4 (OBC Temperature, LoRa)**	RHN	60
21		15:00	15:30	Break			30
22		15:30	17:00	[Lab] Cubesat hands on (Solar Panel)	**Slides Day4 (EPS temperature, battery voltage TMs)**	RHN	90
23	3	9:00	14:30	Asia Tech X @ Expo (Self Explore)	Sign in at SF Booth (else considered no attendance)		330
24		**14:30**	**16:00**	**Asia Tech X @ Expo (Engagement @ Meeting Room)**	Sign in at Meeting Room (else considered no attendance)	RHN, NT	90
25		16:00	17:00	Asia Tech X @ Expo (Self Explore)			60

Agenda

S/N	Day	Time		Title	2nd Space Slides	Trainer	Duration
		Start	End				
26	4	9:00	10:30	Space Environment & Orbital Mechanics	L2 Orbits and Space Environment	RHN	90
27		10:30	11:00	[Lab] Space Mission Design (GMAT)	L14 Space Mission Design SW [GMAT]	RHN, NT	30
28		11:00	11:15	Break			15
29		11:15	12:00	Attitude Determination & Control Subsystem (ADCS)	L8 Attitude Determination and Control System	NT	45
30		12:00	12:45	Propulsion Control Subsystem (PCS)	L12 Satellite propulsion	NT	45
31		12:45	13:45	Lunch			60
32		13:45	15:15	[Lab] Cubesat hands on (TT&C)	**Slides Day5 (VBatt, Solar Panel, TM real-time plotter)**	RHN, NT	90
33		15:15	15:45	Break			30
34		15:45	17:15	[Lab] Cubesat hands on (camera)	**Slides Day5 (, camera)**	RHN, NT	90

Agenda

S/N	Day	Time		Title	2nd Space Slides	Trainer	Duration
		Start	End				
35	5	9:00	10:30	Telemetry, Tracking and Command Subsystem (TT&C)	L7 Satellite Telecommunication System	RHN	90
36		10:30	11:15	Assembly, Integration & Testing (AIT)	L10 Satellite AIT and qualification part 1	NT	45
37		11:15	11:30	Break			15
38		11:30	12:15	LEOP: Launcher, deployers and separators, IOT	L15 Launch systems and space logistic; Cubesat 101	NT	45
39		12:15	13:15	Lunch			60
40		13:15	13:45	Project Management & Quality Assurance	L13 Project and quality management ECSS and Guidelines	NT	30
41		13:45	14:15	Mission Management	L18 Mission management development	NT	30
42		14:15	14:30	Briefing on Project Report on Mission Design	L20 Mission Design – lets design a satellite mission	NT	15
43		14:30	15:00	Break			30
44		15:00	17:00	[Lab] Cubesat hands on (Reaction wheel test)	**Slides Day6 (reaction wheel)**	NT	120
45		17:00	17:15	[Lab] Disassemble Cubesat and Ground Station	reverse the assembly procedure (Slides Day2, Day3)	NT	15
46		17:15	17:45	Certification presentation	Survey, Announcement of SB Proj	LamLap	30

Course Materials

https://github.com/sbwspacefaculty-png/SatBuildWorkshop

Notes

1. Handling of soldered cables, solar panels
2. Explain the circuit of RBF
3. Explain the need to be careful in flashing correct code
4. No sticker/tape on solar panels
5. Follow the Arduino installation instruction correctly
6. Follow the assembly slides correctly

Misc

- 75% attendance for Certificate of Completion
- Final Mission Exercise Report (to submit within 2 weeks)
  - Email to : learn@spacefaculty.asia
- Apply for participation in Satellite Building Project:
  - Email with resume: learn@spacefaculty.asia
  - State the period of availability

Course Materials

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Workshop Survey

EPS Architecture and RBF